Testing notes

Tested on November 11, 2026 with PostgreSQL 17.11.

The tables and sample data loaded in two empty databases. Both runs passed all 42 checks and all 8 example queries. The manager foreign key also rejected an invalid transaction at commit and rolled back the whole transaction.

Full output: test-results.txt.

Run the checks

```
createdb cs166_project
psql -X -v ON_ERROR_STOP=1 -d cs166_project -f load.sql
psql -X -v ON_ERROR_STOP=1 -d cs166_project -f support/test_constraints.sql
psql -X -v ON_ERROR_STOP=1 -d cs166_project -f support/queries.sql
```

Use another name if the database already exists. The tests roll back their changes, so the sample data stays the same.

Sample data

Table	Rows
professor	3
course	3
department	2
worksin	4
student	6
teach	5
assignment	11
enroll	12
assign	17

Sample rows use fictional names and fixed dates/scores, entered directly in `sample_data.sql`. They cover three terms, repeated sections, multiple department memberships, late submissions, missing and ungraded work, tied scores, and grades of 0 and 100. Each sample section has 100 total weight.

What was checked

- Duplicate keys, missing required values, and invalid names or numbers fail.
- Grades outside 0-100 and weights outside 0-100 fail. Weight cannot be zero.
- Foreign keys reject missing parents and scores for unenrolled students.
- Managers and section professors must work in the correct department.
- Referenced records cannot be deleted.
- Valid inserts, updates, and deletes work.
- Terms keep repeated sections separate. One student can have multiple grades in the same section.
- Zero counts as a grade. NULL means ungraded.
- Totals of 90 or 110 do not produce a final grade.
- Failed transactions and the test script leave the sample data unchanged.

Example results

Query	Result
Student 101, Fall CS166 current grade	86.00%
Average needed on remaining work to finish with 70%	54.00%
Final projection with 90% on remaining work	88.00%
Spring CS166 final grades	101: 84%; 102: 94%; 103: 66%
Fall CS166 students with no grades	104 and 106
Fall CS166 assignment 1 min/max/average	0% / 100% / 60%
GPA	101: 2.50; 102: 4.00; 103: 1.00
Professor 1 average by term	2025-Fall: 75%; 2026-Spring: 81.33%

The SQL was tested locally, not on the course server. GUI testing, permissions, and index performance testing are for Phase 2.